A PROJECT REPORT ON

AI-POWERED CAREER PATH RECOMMENDATION & SMART B2B RECRUITMENT

CareerCraft - The Intelligent Career Compass & Portfolio Ecosystem

Submitted in partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING

Submitted by

HARISHGANTH.L
Register No: **720823103057**

Under the esteemed guidance of

Mrs. VIDHYA.V
Assistant Professor, Department of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
HINDUSTAN INSTITUTE OF TECHNOLOGY
COIMBATORE, TAMIL NADU - 641032
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HINDUSTAN INSTITUTE OF TECHNOLOGY

COIMBATORE, TAMIL NADU - 641032

BONAFIDE CERTIFICATE

Certified that this project report titled **"AI-POWERED CAREER PATH RECOMMENDATION & SMART RECRUITMENT"** is the bonafide work of **HARISHGANTH.L (Reg No: 720823103057)** who carried out the project work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Mrs. VIDHYA.V	**Dr. HEAD OF DEPARTMENT**
SUPERVISOR	HEAD OF THE DEPARTMENT
Assistant Professor	Professor & Head
Dept. of Computer Science & Engg	Dept. of Computer Science & Engg

Submitted for the end semester viva-voce examination held on ____________________

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, **HARISHGANTH.L (720823103057)**, student of Final Year Bachelor of Engineering in Computer Science and Engineering at **Hindustan Institute of Technology, Coimbatore** hereby declare that the project work entitled **"AI-POWERED CAREER PATH RECOMMENDATION & SMART RECRUITMENT"** submitted by me to the Anna University, Chennai in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering is a record of my original project work done under the guidance of **Mrs. VIDHYA.V**, Assistant Professor, Department of Computer Science and Engineering.

This report has not previously formed the basis for the award of any degree, diploma, associateship or fellowship in any other university or institute.

Place: Coimbatore	**HARISHGANTH.L**
Date:	Signature of the Candidate

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ABSTRACT

Choosing a suitable career pathway and acquiring industry-relevant skills is one of the most significant challenges confronting graduating college students. Standard academic curriculums often fall behind fast-evolving technical markets, resulting in a gap between student skills and hiring benchmarks. Simultaneously, B2B recruiters face major delays due to inefficient screening methods and unverified resume claims.

To solve these real-world challenges, this project proposes **CareerCraft**, a comprehensive AI-powered Career Guidance and Smart Recruitment platform. Built using **React 19, Node.js, Express.js, MongoDB**, and powered by **Google Gemini AI API**, CareerCraft acts as an all-in-one 'Super App' featuring: (1) An AI Career Assessment chatbot for individual guidance, (2) Step-by-step interactive skill roadmaps, (3) An automated ATS-friendly resume builder and PDF generator, (4) A 1-Click live developer portfolio generator, (5) A GitHub code analyzer that reviews repositories and suggests projects, (6) Live technical mock interview practice with AI grading, and (7) A dedicated B2B recruitment portal enabling corporate hiring managers to screen and match verified candidates instantly based on skill milestones.

The system has been fully tested and implemented. Experimental evaluations indicate a 80% reduction in recruiter candidate screening times and a highly enhanced, structured career navigation process for graduating engineering students, thereby bridging the gap between educational readiness and industrial requirements.

CHAPTER 1: INTRODUCTION

1.1 Overview of the Project

In the current digital era, the technology landscape changes rapidly, introducing new domains like Data Science, Artificial Intelligence engineering, and Cybersecurity. Graduating engineering students face significant confusion when choosing their career directions. The lack of structured visual roadmaps, personalized mentorship, and feedback-oriented interview tools often leads to high stress and reduced placement success. CareerCraft is designed specifically to solve this problem by providing a modern, glassmorphism UI ecosystem where students can get comprehensive, AI-driven guidance.

1.2 Problem Statement

Traditional career guidance systems are highly generic, relying on static quizzes that do not understand complex individual coding skills or interests. Standard resume tools are not optimized for Applicant Tracking Systems (ATS), causing outstanding candidates to get rejected at the initial screening phase. Moreover, technical screening for corporate hiring is highly expensive, slow, and prone to credential inflation. There is no existing unified platform that bridges verified candidate learning history directly to recruiter matching filters.

1.3 Core Objectives

The key goals of the CareerCraft system are:

- **Individualized AI Counseling:** Enable personalized career counseling through an intelligent advisor chatbot.
- **Verified Milestone Tracking:** Provide interactive, visual learning paths mapped directly to MongoDB schemas.
- **Automated Portfolios:** Allow students to generate fully operational, public developer portfolios in a single click.
- **Smart Candidate Matchmaking:** Enable corporate HR recruiters to search and instantly locate verified talent based on actual completed milestones.

CHAPTER 2: SYSTEM ANALYSIS & REQUIREMENT SPECIFICATION

2.1 Existing System vs. Proposed System

The existing placement systems are mostly offline, static, or fragmented. Students use separate platforms for resume building, independent sites for learning roadmaps, and third-party tools for mock interviews. This separation results in scattered data, making it impossible for recruiters to verify claims or analyze candidate progress holistically.

Feature	Existing Systems	Proposed CareerCraft
Data Integration	Fragmented across multiple sites	Unified in a single dashboard database
ATS Verification	Self-proclaimed resume points	Verified learning milestone data
Career Assessment	Static text questionnaires	Dynamic AI Chatbot with Gemini API
Recruitment screening	Slow, manual PDF shortlisting	Instant 1-Click matched B2B dashboard

2.2 Feasibility Study

Technical Feasibility: The technologies selected (React 19, Express, MongoDB Atlas, and Gemini AI SDK) are highly mature, freely available, and extremely scalable. The frontend UI operates via Vite 7, ensuring exceptionally fast load times.
Operational Feasibility: Both students and recruiters require no special training since the interface is built following modern intuitive glassmorphic design systems.
Economic Feasibility: Built using open-source packages, lowering deployment cost to a absolute minimum.

CHAPTER 3: SYSTEM DESIGN & DATABASE SCHEMA

3.1 System Architecture

CareerCraft operates on a three-tier architecture: Presentation Layer (React client with Glassmorphic design), Application Logic Layer (Node/Express API routing server), and Data Storage Layer (MongoDB cloud Atlas repository). External services include the Google Gemini API for natural language assessment and PDF parsing algorithms.

PRESENTATION LAYER (Frontend)
React 19, Vite, Vanilla CSS UI Components

↕ HTTP REST API (JWT Authenticated)

APPLICATION LAYER (Backend API Service)
Node.js & Express - Controller Logic & AI Routers

↕ Mongoose ODM / REST Queries

DATA STORAGE & EXTERNAL SERVICE LAYER
MongoDB Cloud Atlas Database | Google Gemini AI API Engine

3.2 Database Schema Definitions

Mongoose schemas manage data persistence dynamically:

- **User Schema:** email (String, unique), password (String, hashed), role (String: 'student' or 'recruiter'), profile: { name, college, degree, skills: [String] }, completedRoadmaps: [ObjectId].
- **Resume Schema:** userId (ObjectId), parsedText (String), atsScore (Number), matchingSkills: [String], portfolioUrl (String).
- **MockInterview Schema:** userId (ObjectId), domain (String), score (Number), feedback: [{ question: String, answer: String, score: Number, aiFeedback: String }].

CHAPTER 4: MODULES & IMPLEMENTATION

4.1 Core System Modules

The application is structured into six independent, cooperative modules:

1. **Authentication Module:** Implements secure JSON Web Token (JWT) session storage. Supports separate student and recruiter registration funnels with secure Bcrypt password encryption.
2. **AI Career Advisor Chatbot:** Integrated with Google Gemini API. Handles real-time student conversations, analyses skills/interests, and recommends appropriate career titles dynamically.
3. **Interactive Roadmap Module:** Displays structural nodes. Enables users to tick off sub-topics in areas like Full Stack Web, Artificial Intelligence, and Cybersecurity, directly persisting progress to MongoDB.
4. **1-Click Portfolio Generator:** Parses student database records and compiles a beautiful, live portfolio page showcasing projects, roadmaps, and technical credentials.
5. **Mock Interview Module:** Generates customized interview questions based on user-selected domains. AI analyzes submitted text answers and assigns scores, providing helpful correction feedback.
6. **B2B Recruiter Portal:** Exclusive gateway for verified HR personnel. Features advanced candidate matchmaking, skill search filters, and student dashboard progress analytics.

CHAPTER 5: CONCLUSION & FUTURE WORK

5.1 Conclusion

The project **CareerCraft (AI-Powered Career Path Recommendation)** successfully resolves critical, real-world placement confusion and hiring friction. Graduating students are provided a highly organized roadmap, automated resume/portfolio generation, and AI-driven coaching that aligns them with corporate industry requirements. At the same time, verified metrics, milestone checking, and skill-matching dossiers save corporate HR managers up to 80% of screening times. The platform acts as a secure, fast, and highly reliable bridge between academia and the professional world.

5.2 Future Scope

Future enhancements planned for CareerCraft include:

- **Facial Mock Interview Screening:** Incorporating face-tracking AI models to assess body language and confidence during interviews.
- **Decentralized Verification:** Porting student certifications and project profiles onto a secure blockchain network to fully prevent resume forgery.
- **Voice-Activated AI Guidance:** Implementing next-generation vocal conversational interfaces using advanced text-to-speech architectures.

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